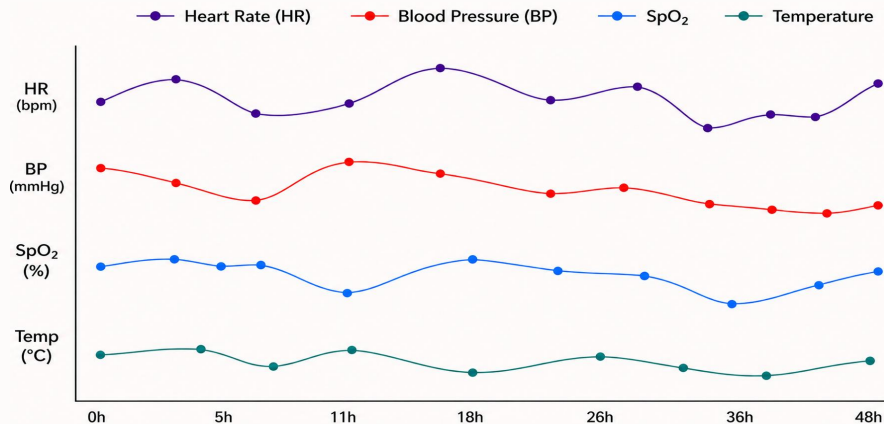


(A) Multi-Scale Temporal Interface

Irregular Temporal Input



Text Input

In first 48 hours, compare the first HR decrease with the first temperature decrease ...

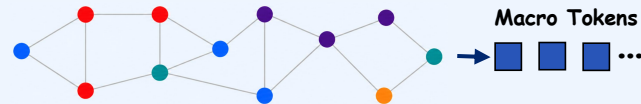
Text Encoder

Tokens

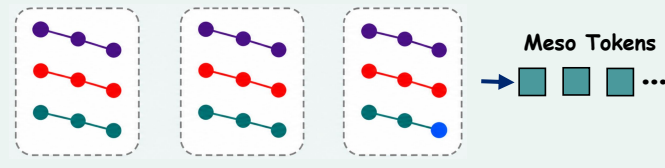
Text Embedding (caption encoding)

Irregularity-Aware Multi-Scale Encoder

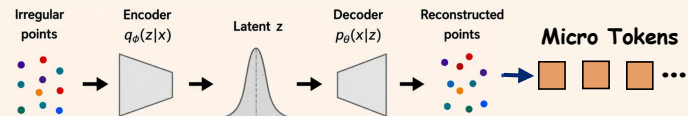
Macro-Scale Encoder (E_g)



Meso-Scale Encoder (E_p)



Micro-Scale Encoder (E_m)



Temporal Evidence Distiller

Scale Projection

Macro Projection

Meso Projection

Micro Projection

Hierarchical Fusion

Query-Based Resampler

Compressed Time-Series Tokens

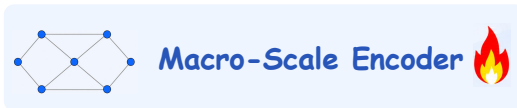
Join with Text Tokens

Large Language Model

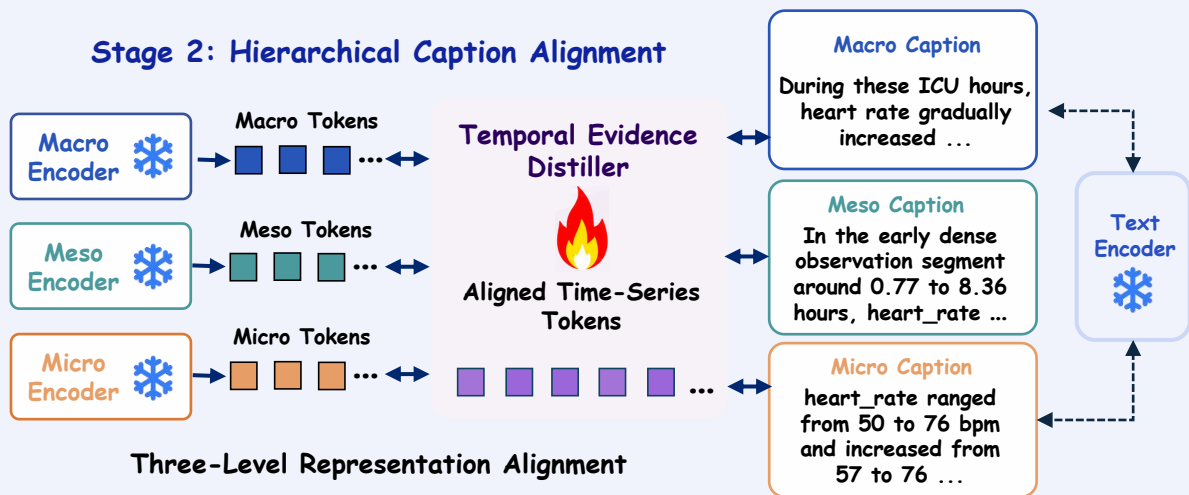
Answer

(B) Progressive Temporal-Language Alignment

Stage 1: Multi-Scale Encoder Pretraining



Stage 2: Hierarchical Caption Alignment



Stage 3: QA-Oriented Adaptation

Step 1: QA-Oriented Distiller Tuning



Step 2: Joint LoRA-Distiller Adaptation

